

Deuterium-Mediated Neutron Chain Fusion Initiated by Proton–Electron Capture with Lithium-6 Yield Enhancement: A Tritium Breeding Mechanism for Existing Tokamak Infrastructure

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Abstract

We propose a theoretical mechanism in which free protons injected into dense deuterium gas undergo Coulomb-barrier-free electron capture ($p + e^- \rightarrow n + \nu_e$), producing neutrons that subsequently convert deuterium nuclei into tritium via $n + {}^2\text{H} \rightarrow {}^3\text{H} + p + 2.22\text{ MeV}$. The liberated proton re-enters the chain, sustaining neutron production. Accumulated tritium and deuterium then undergo fusion (${}^3\text{H} + {}^2\text{H} \rightarrow {}^4\text{He} + n + 17.6\text{ MeV}$), with Lithium-6 acting as a yield enhancer by capturing residual neutrons ($n + {}^6\text{Li} \rightarrow {}^4\text{He} + {}^3\text{H} + 4.78\text{ MeV}$) and recycling tritium back into the fusion cycle. The central contribution of this work is that the proton–electron capture chain provides a self-sustaining *in situ* tritium breeding mechanism fully compatible with existing tokamak infrastructure, addressing the critical tritium supply problem in D–T fusion programs.

Keywords: tritium breeding, deuterium, proton-electron capture, neutron chain, tokamak, ITER, lithium-6, D–T fusion

1 Introduction

Deuterium–tritium (D–T) fusion is the primary reaction pursued by major tokamak programs worldwide, including ITER and China’s EAST reactor [?, ?]. The reaction:



releases 17.6 MeV per event and is the most accessible fusion reaction at achievable plasma temperatures ($T \sim 10^8\text{ K}$). However, tritium (${}^3\text{H}$) does not occur naturally in significant quantities and has a half-life of only 12.3 years, making tritium supply one of the central unsolved problems of fusion energy [?].

Current tritium breeding concepts rely on neutron bombardment of lithium blankets surrounding the reactor. While viable in principle, this approach requires careful neutron

economy management and does not achieve tritium self-sufficiency without a breeding ratio exceeding unity [?].

In this paper, we propose a complementary mechanism: a proton–electron capture chain in deuterium gas that produces neutrons without a Coulomb barrier, converts deuterium to tritium *in situ*, and feeds the resulting tritium directly into the D–T fusion cycle. Lithium-6 serves as a yield enhancer rather than the primary energy source, capturing excess neutrons and regenerating tritium.

2 Reaction Mechanism

2.1 Step 1: Coulomb-Barrier-Free Neutron Production

A free proton with kinetic energy $E_k \geq 1.293$ MeV is injected into dense deuterium gas. The proton is attracted to the bound electron of a deuterium atom (opposite charges, no Coulomb barrier):

$$p + e^- \rightarrow n + \nu_e \quad \Delta mc^2 = -1.293 \text{ MeV} \quad (2)$$

This is identical to the mechanism proposed in our companion paper [?], extended here to a deuterium medium.

2.2 Step 2: Neutron–Deuterium Tritium Breeding

The produced neutron interacts with a deuterium nucleus:

$$n + {}^2\text{H} \rightarrow {}^3\text{H} + p + 2.22 \text{ MeV} \quad (3)$$

This reaction has **no Coulomb barrier** (neutron is neutral), proceeds with thermal neutrons, and has a well-measured cross-section of $\sigma \approx 0.52$ mb at thermal energies [?]. The liberated proton (p) re-enters the chain at Step 1, sustaining neutron production without external proton injection after initiation.

2.3 Step 3: D–T Fusion

Accumulated tritium undergoes fusion with deuterium:

$${}^3\text{H} + {}^2\text{H} \rightarrow {}^4\text{He} + n + 17.6 \text{ MeV} \quad (4)$$

The Coulomb barrier for this step is overcome by the same methods employed in existing tokamak reactors: magnetic confinement and plasma heating to $T \sim 10^8$ K. **No new technology is required for this step.** The novelty of this proposal lies in providing the tritium fuel via Steps 1–2.

2.4 Step 4: Lithium-6 Yield Enhancement

Neutrons produced in Eq. (??) are captured by a Lithium-6 blanket:

$$n + {}^6\text{Li} \rightarrow {}^4\text{He} + {}^3\text{H} + 4.78 \text{ MeV} \quad (5)$$

The tritium produced here is recycled back into Step 3, increasing the tritium breeding ratio. Lithium-6 is thus a *yield enhancer*, not the primary energy source.

3 Complete Reaction Cycle

The full cycle is:



Step S2 regenerates the proton for S1, closing the neutron production loop. Step S4 regenerates tritium for S3, closing the fusion fuel loop.

3.1 Net Energy per Full Cycle

Table 1: Energy budget per complete reaction cycle

Process	Energy	Direction
$p + e^- \rightarrow n + \nu_e$	1.293 MeV	Input
$n + {}^2\text{H} \rightarrow {}^3\text{H} + p$	2.220 MeV	Output
${}^3\text{H} + {}^2\text{H} \rightarrow {}^4\text{He} + n$	17.600 MeV	Output
$n + {}^6\text{Li} \rightarrow {}^4\text{He} + {}^3\text{H}$	4.780 MeV	Output
Net per cycle	23.307 MeV	Output

$$\boxed{E_{\text{net}} = 2.220 + 17.600 + 4.780 - 1.293 = 23.307 \text{ MeV per cycle}} \quad (6)$$

4 Comparison with Existing Approaches

Table 2: Comparison with existing tritium breeding and fusion approaches

Feature	ITER Blanket	Standard D-T	This Work
Tritium source	Li blanket	External supply	In situ chain
Coulomb barrier	N/A	Yes (plasma)	None (S1, S2)
Neutron source	Fusion	Fusion	p+e chain
Li-6 role	Primary breed	None	Yield enhancer
Self-sustaining	Partial	No	Yes (loop)
Net energy (MeV)	17.6	17.6	23.307
New infrastructure	Required	Required	None needed

The key advantage over ITER’s blanket breeding approach is that the proton–electron capture chain produces tritium *before* fusion occurs, rather than relying on fusion neutrons to breed tritium after the fact. This eliminates the startup tritium inventory problem.

5 Compatibility with Existing Tokamaks

The proposed mechanism requires no modification to existing tokamak reactor designs. The deuterium gas chamber for Steps S1–S2 can be implemented as a pre-ionization stage upstream of the plasma chamber. The Lithium-6 blanket (Step S4) is already a planned component of ITER and successor reactors.

The only addition is a proton beam source ($E_k \geq 1.293$ MeV) to initiate the chain. Once initiated, Step S2 regenerates protons internally, making the chain self-sustaining in terms of fuel.

6 Open Challenges

1. **Chain initiation energy.** The initial proton beam requires $E_k \geq 1.293$ MeV. Standard particle accelerators provide this trivially.
2. **Neutron thermalization.** The neutron produced in S1 must be thermalized before interacting with deuterium in S2, as the cross-section for $n + {}^2\text{H}$ is highest at thermal energies. A deuterium moderator layer achieves this naturally.
3. **Tritium confinement.** Tritium produced in S2 must be confined and directed into the plasma chamber. This is a known engineering challenge already addressed in ITER design documents.
4. **Neutrino energy loss.** Each S1 event loses ~ 0.511 MeV via neutrino emission. This is included in the energy balance above.

7 Proposed Experimental Test

1. Fill a high-pressure deuterium chamber ($P > 50$ atm) surrounded by neutron detectors and a Lithium-6 blanket.
2. Inject a proton beam at $E_k = 5\text{--}20$ MeV.
3. Measure tritium production rate as a function of deuterium pressure. A superlinear dependence would indicate chain propagation beyond single-collision production.
4. Measure heat output in the Li-6 blanket via calorimetry.
5. If tritium production is confirmed, feed accumulated tritium into a small tokamak or magnetic mirror device and measure D–T fusion yield.

8 Conclusion

We have proposed a deuterium-based neutron chain mechanism that produces tritium *in situ* via proton–electron capture, without requiring a Coulomb barrier for the neutron production and breeding steps. The key contributions are:

- A self-sustaining proton–neutron–tritium production loop in dense deuterium gas.

- Net energy output of 23.307 MeV per complete cycle, compared to 17.6 MeV for standard D–T fusion alone.
- Full compatibility with existing tokamak infrastructure — no new reactor design required.
- Lithium-6 as a tritium recycler and yield enhancer, increasing the effective breeding ratio beyond unity.

If experimentally confirmed, this mechanism could resolve the tritium supply bottleneck that currently limits the timeline for commercial fusion energy.

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